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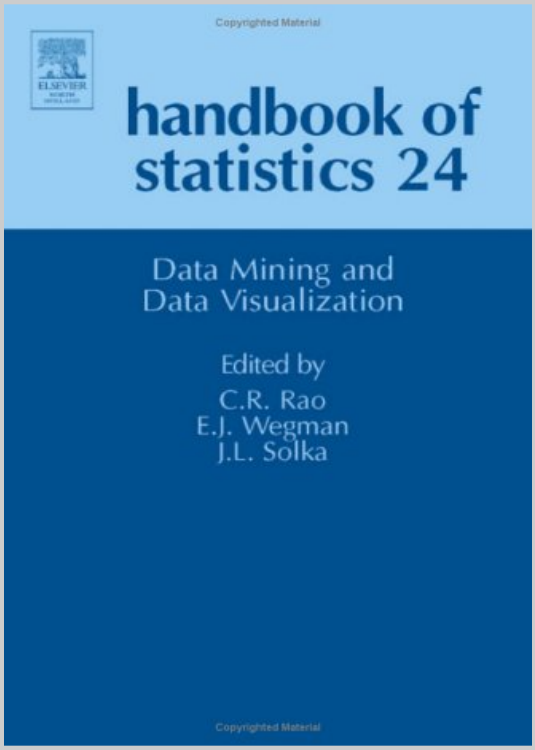
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Preface

It has long been a philosophical theme that statisticians ought to be data centric as opposed to methodology centric. Throughout the history of the statistical discipline, the most innovative methodological advances have come when brilliant individuals have wrestled with new data structures. Inferential statistics, linear models, sequential analysis, nonparametric statistics, robust statistical methods, and exploratory data analysis have all come about by a focus on a puzzling new data structure. The computer revolution has brought forth a myriad of new data structures for researchers to contend with including massive datasets, high-dimensional datasets, opportunistically collected datasets, image data, text data, genomic and proteomic data, and a host of other data challenges that could not be dealt with without modern computing resources.

This volume presents a collection of chapters that focus on data; in our words, it is data-centric. Data mining and data visualization are both attempts to handle nonstandard statistical data, that is, data, which do not satisfy traditional assumptions of independence, stationarity, identically distribution, or parametric formulations. We believe it is desirable for statisticians to embrace such data and bring innovative perspectives to these emerging data types.

This volume is conceptually divided into three sections. The first focuses on aspects of data mining, the second on statistical and related analytical methods applicable to data mining, and the third on data visualization methods appropriate to data mining. In Chapter 1, Wegman and Solka present an overview of data mining including both statistical and computer science-based perspectives. We call attention to their description of the emerging field of massive streaming datasets. Kaufman and Michalski approach data mining from a machine learning perspective and emphasize computational intelligence and knowledge mining. Marchette describes exciting methods for mining computer security data with the important application to cybersecurity. Martinez turns our attention to mining of text data and some approaches to feature extraction from text data. Solka et al. also focuses on text mining applying these methods to cross corpus discovery. They describe methods and software for discovery subtle, but significant associations between two corpora covering disparate fields. Finally Duric et al. round out the data mining methods with a discussion of information hiding known as steganography.

The second section, on statistical methods and related methods applicable to data mining begins with Rao's description of methods applicable to dimension reduction and graphical representation. Hand presents an overview of methods of statistical pattern recognition, while Scott and Sain present an update of Scott's seminal 1992 book on multivariate density estimation. Hubert et al., in turn, describe the difficult problem

of analytically determining multivariate outliers and their impact on robustness. Sutton describes recent developments in classification and regression trees, especially the concepts of bagging and boosting. Marchette et al. describe some new computationally effective classification tools, and, finally Said gives an overview of genetic algorithms.

The final section on data visualization begins with a description of rotations (grand-tour methods) for high-dimensional visualization by Buja et al. This is followed by Carr's description of templates and software for showing statistical summaries, perhaps the most novel of current approaches to visual data presentation. Wilhelm describes in depth a framework for interactive statistical graphics. Finally, Chen describes a computer scientist's approach to data visualization coupled with virtual reality.

The editors sincerely hope that this combination of philosophical approaches and technical descriptions stimulates, and perhaps even irritates, our readers to encourage them to think deeply and with innovation about these emerging data structures and develop even better approaches to enrich our discipline.

C.R. Rao
E.J. Wegman
J.L. Solka

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Handbook of Statistics, Vol. 24
ISSN: 0169-7161
2005 Published by Elsevier B.V.
DOI 10.1016/S0169-7161(04)24001-9

Statistical Data Mining

Edward J. Wegman and Jeffrey L. Solka

Abstract

This paper provides an overview of data mining methodologies. We have been careful during our exposition to focus on many of the recent breakthroughs that might not be known to the majority of the community. We have also attempted to provide a brief overview of some of the techniques that we feel are particularly beneficial to the data mining process. Our exposition runs the gambit from algorithmic complexity considerations, to data preparation, databases, pattern recognition, clustering, and the relationship between statistical pattern recognition and artificial neural systems.

Keywords: dimensionality reduction; algorithmic complexity; statistical pattern recognition; clustering; artificial neural systems

1. Introduction

The phrases ‘data mining’, and, in particular, ‘statistical data mining’, have been at once a pariah for statisticians and also a darling. For many classically trained statisticians, data mining has meant the abandonment of the probabilistic roots of statistical analysis. Indeed, this is exactly the case simply because the datasets to which data mining techniques are typically applied are opportunistically acquired and were meant originally for some other purpose, for example, administrative records or inventory control. These datasets are typically not collected according to widely accepted random sampling schemes and hence inferences to general situations from specific datasets are not valid in the usual statistical sense. Nonetheless, data mining techniques have proven their value in the marketplace. On the other hand, there has been considerable interest in the statistics community in recent years for approaches to analyzing this new data paradigm.

The landmark paper of Tukey (1962), entitled “The future of data analysis,” and later in the book, *Exploratory Data Analysis*, John Tukey (1977) sets forth a new paradigm for statistical analysis. In contrast to what has come to be called confirmatory analysis in which a statistical model is assumed and inference is made on the parameters of that model, exploratory data analysis (EDA) is predicated on the fact that we do not

necessarily know that model assumptions actually hold for data under investigation. Because the data may not conform to the assumptions of the confirmatory analysis, inferences made with invalid model assumptions are subject to (potentially gross) errors. The idea then is to explore the data to verify that the model assumptions actually hold for the data in hand. It is a very short leap of logic to use exploratory techniques to discover unanticipated structure in the data. With the rise of powerful personal computing, this more aggressive form of EDA has come into vogue. EDA is no longer used to simply verify underlying model assumptions, but also to uncover unanticipated structure in the data.

Within the last decade, computer scientists operating in the framework of databases and information systems have similarly come to the conclusion that a more powerful form of data analysis could be used to exploit data residing in databases. That work has been formulated as knowledge discovery in databases (KDD) and data mining. A landmark book in this area is (Fayyad et al., 1996). The convergence of EDA from the statistical community and KDD from the computer science community has given rise to a rich if somewhat tense collaboration widely recognized as data mining.

There are many definitions of data mining. The one we prefer was given in (Wegman, 2003). Data mining is an extension of exploratory data analysis and has basically the same goals, the discovery of unknown and unanticipated structure in the data. The chief distinction lies in the size and dimensionality of the data sets involved. Data mining, in general, deals with much more massive data sets for which highly interactive analysis is not fully feasible.

The implication of the distinction mentioned above is the sheer size and dimensionality of data sets. Because scalability to massive datasets is one of the cornerstones of the data mining activity, it is worthwhile for us to begin our discussion of statistical data mining with a discussion of the taxonomy of data set sizes and their implications for scalability and algorithmic complexity.

2. Computational complexity

2.1. Order of magnitude considerations

In Table 1 we present Peter Huber's taxonomy of data set complexity (Huber, 1992, 1994). This was expanded by Wegman (1995). This taxonomy provides a feel for the magnitudes of the datasets that one is often interested in analyzing. We present a descriptor for the dataset, the size of the dataset in bytes, and the appropriate storage mode for such a dataset. Originally, the statistics community was often concerned with datasets that resided in the Small to Medium classes. Numerous application areas such as computer security and text processing have helped push the community into analyzing datasets that reside in the Large or Huge classes. Ultimately we will all be interested in the analysis of datasets in the Massive or Supermassive class. Such monstrous datasets will necessitate the development of recursive methodologies and will require the use of those approaches that are usually associated with streaming data.

Table 1
Huber–Wegman taxonomy of data set sizes

Descriptor	Data set size in bytes	Storage mode
Tiny	10^2	Piece of paper
Small	10^4	A few pieces of paper
Medium	10^6	A floppy disk
Large	10^8	Hard disk
Huge	10^{10}	Multiple hard disks, e.g. RAID storage
Massive	10^{12}	Disk farms/tape storage silos
Supermassive	10^{15}	Distributed data centers

Table 2
Algorithmic complexity

Complexity	Algorithm
$O(r)$	Plot a scatterplot
$O(n)$	Calculate means, variances, kernel density estimates
$O(n \log(n))$	Calculate fast Fourier transforms
$O(nc)$	Calculate singular value decomposition of an rc matrix; solve a multiple linear regression
$O(nr), O(n^{3/2})$	Solve a clustering algorithm with $r \propto \text{sqrt}(n)$
$O(n^2)$	Solve a clustering algorithm with c fixed and small so that $r \propto n$

Let us consider a data matrix consisting of r rows and c columns. One can calculate the total number of entries n as $n = rc$. In the case of higher-dimensional data, we write $d = c$ and refer to the data as d-dimensional. There are numerous types of operations or algorithms that we would like to utilize as part of the data mining process. We would like to be able to plot our data as a scatterplot, to compute summary statistics for our data such as means and variances, to perform probability density estimation on our data set using the standard kernel density estimator approach, we might wish to apply the fast Fourier transform to our dataset, we would like to be able to obtain the singular value decomposition of a multi-dimensional data set in order to ascertain appropriate linear subspaces that capture the nature of the data, we may wish to perform a multiple linear regression on a multi-dimensional dataset, we may wish to applying a clustering methodology with r proportional to sqrt(n), or, finally, we might wish to solve a clustering algorithm with c fixed and small so that r is proportional to n. This list of algorithms/data analysis techniques is not exhaustive but the list does represent many of the tasks that one may need to do as part of the data mining process. In Table 2 we examine algorithmic complexity as a function of these various statistical/data mining algorithms.

Now it is interesting to match the computational requirements of these various algorithms against the various different dataset sizes that we discussed previously. This allows one to ascertain the necessary computational resources in order to have a hope of applying each of the particular algorithms to the various datasets. Table 3 details the number of operations (within a constant multiplicative factor) for algorithms of various computational complexities and various data set sizes.

Table 3
Number of operations for algorithms of various computational complexities and various data set sizes

n	$n^{1/2}$	n	$n\log(n)$	$n^{3/2}$	n^2
Tiny	10	10^2	2×10^2	10^3	10^4
Small	10^2	10^4	4×10^4	10^6	10^8
Medium	10^3	10^6	6×10^6	10^9	10^{12}
Large	10^4	10^8	8×10^8	10^{12}	10^{16}
Huge	10^5	10^{10}	10^{11}	10^{15}	10^{20}
Massive	10^6	10^{12}	1.2×10^{13}	10^{18}	10^{24}
Supermassive	$10^{7.5}$	10^{15}	1.5×10^{16}	$10^{22.5}$	10^{30}

2.2. *Feasibility limits due to CPU performance*

Next we would like to take these computational performance figures of merit and examine the possibility of executing them on some current hardware. It is difficult to know exactly which machine should be included in the list of current hardware since the machine performance capabilities are constantly changing as the push for faster CPU speeds continues. We will use 4 machines for our computational feasibility analysis. The first machine will be a 1 GHz Pentium IV (PIV) machine with a sustainable performance of 1 gigaflop. The second machine will consist of a hypothetical flat neighborhood network of 12 1.4 GHz Athlons with a sustainable performance of 24 gigaflops. The third machine will be a hypothetical flat neighborhood network of 64 0.7 GHz Athlons with a sustained performance of 64 gigaflops. The fourth and final machine will be a hypothetical massively distributed grid type architecture with a sustained processing speed of 1 teraflop. This list of machines runs the gambit from a relatively low end system to a state of the art high performance "super computer".

In Table 4 we present execution times for a hypothetical 1 GHz PIV machine with a sustained performance level of 1 gigaflop when applied to the various algorithm/dataset combinations. In Table 5 we present execution times for a hypothetical Flat Neighborhood Network of 12 1.4 GHz Athlons with a sustained performance of 24 gigaflops when applied to the various algorithm/dataset combinations. In Table 6 we present execution times for a hypothetical Flat Neighborhood Network of 64 0.7 GHz Athlons with a sustained performance of 64 gigaflops. And finally in Table 7 we present mini-

Table 4
Execution speed of the various algorithm/dataset combinations on a Pentium IV 1 GHz machine with 1 gigaflop performance assumed

n	$n^{1/2}$	n	$n\log(n)$	$n^{3/2}$	n^2
Tiny	10^{-8} s	10^{-7} s	2×10^{-7} s	10^{-6} s	10^{-5} s
Small	10^{-7} s	10^{-5} s	4×10^{-5} s	0.001 s	0.1 s
Medium	10^{-6} s	0.001 s	0.006 s	1.002 s	16.74 min
Large	10^{-5} s	0.1 s	0.78 s	16.74 min	115.7 days
Huge	10^{-4} s	10.02 s	1.668 min	11.57 days	3170 years

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